FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Main)-2020

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

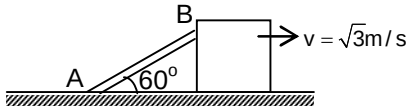
Physics

PART – I

SECTION – A
(One Options Correct Type)

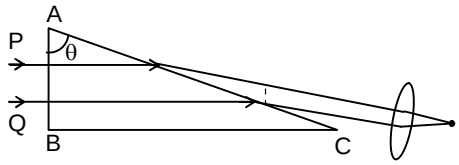
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A rod AB is shown in figure. End A of the rod is fixed on the ground. Block is moving with velocity $\sqrt{3}$ m/s towards right. The speed of end B of the rod when rod makes an angle of 60° with the ground is



(A) 3 m/s (B) 2 m/s
(C) 5 m/s (D) 2.5 m/s

2. Two parallel beams of light P and Q are incident normally on a prism and the transmitted rays are brought to focus with the help of convergent lens as shown in figure. If intensities of the upper and lower beams immediately after transmission from face AC are 4I and I respectively. If resultant intensity at the focus is



(A) I (B) 5I
(C) 17I (D) 9I

3. A thin prism P_1 with angle 4° and made from glass (μ = 1.54) is combined with another prism P_2 made of another glass of μ = 1.72 to produce dispersion without deviation. The angle of prism P_2 is

(A) 53.3^o (B) 4°
(C) 3° (D) 2.6^o

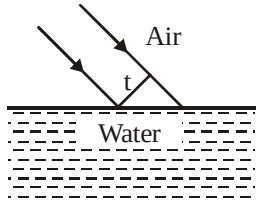
4. A light emitting diode (LED) has a voltage drop of 2V across it and passes a current of 10 mA. When it operates with a 6V battery through a limiting resistor R, the value of R is:

(A) 40 k Ω (B) 4 k Ω
(C) 200 Ω (D) 400 Ω

Space for Rough work

5. A mirror (μ = 3/2) is 10 cm thick. An object is placed 15 cm in front of it. The position of image from the front surface is
(A) 15 cm (B) 21.67 cm
(C) 28.34 cm (D) 35 cm

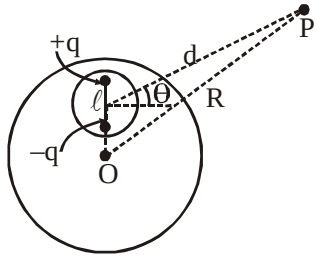
6. A monochromatic beam of width t is incident at 45º on an air water interface as shown in the figure. The refractive index of water is μ and that of air is 1. The width of the beam in water is,



(A) $(\mu - 1)$ T (B) μ T
(C) $\frac{\sqrt{\mu^2 - 1}}{\mu}t$ (D) $\frac{\left(\sqrt{2\mu^2 - 1}\right)}{\mu}t$

7. An object is placed at a distance of 12 cm from a converging lens on its principal axis and a virtual image of certain size is formed. Now the lens is moved 4 cm away from the object and a real image of the same size as that of the (previous) virtual image is formed. The focal length in cm of the lens is:
(A) 14 (B) 16
(C) 18 (D) 20

8. A spherical cavity is created in a neutral solid conducting sphere. Inside the cavity, a dipole is placed as shown in the figure. Electrostatic potential at point P only due to charge induced on the inner surface of the cavity is (assume that $\ell << d$)



(A) $\frac{q\ell}{4\pi\varepsilon_0}\frac{\sin\theta}{d^2}$ (B) $\frac{-q\ell\sin\theta}{4\pi\varepsilon_0 d^2}$
(C) $-\frac{q\ell}{4\pi\varepsilon_0}\frac{\sin\theta}{d}$ (D) $\frac{q\ell\sin\theta}{4\pi\varepsilon_0 d}$

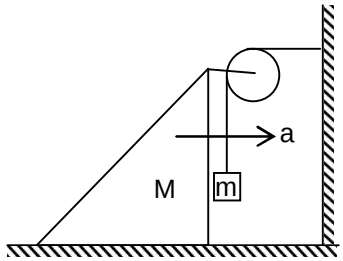
9. The mass of a body is measured as 10.1 kg. The possible percentage error in the measurement is
(A) $\pm$ 1% (B) $\pm$ 0.1%
(C) $\pm$ 10% (D) $\pm$ 0.01%

Space for Rough work

10. If wedge is moving with acceleration a as shown in the figure then value of net force on m is

(A) ma (B) $\sqrt{2}$ ma

(C) mg (D) zero



11. Hydrogen (H), deuerium (D), singly ionised helium $(He)^+$ and doubly ionised (Li^{++}) all have one electron round the nucleus. Consider n = 2 to n = 1 transition. The wavelength of emitted radiations are λ_1, λ_2, λ_3 and λ_4 respectively. Then approximately

(A) $\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$ (B) $4\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

(C) $\lambda_1 = 2\lambda_2 = 2\sqrt{2}\lambda_3 = 3\sqrt{2}\lambda_4$ (D) $\lambda_1 = \lambda_2 = 2\lambda_3 = 3\lambda_4$

12. A tunnel is made across the earth passing through its centre. A ball is dropped from a height h in the tunnel. The motion will be periodic with time period

(A) $2\pi\sqrt{\frac{R}{g}} + \sqrt{\frac{2h}{g}}$ (B) $2\pi\sqrt{\frac{2R}{g}} + 4\sqrt{\frac{h}{g}}$

(C) $2\pi\sqrt{\frac{R}{g}} + 4\sqrt{\frac{2h}{g}}$ (D) $2\pi\sqrt{\frac{2R}{g}} + \sqrt{\frac{h}{g}}$

13. An open organ pipe of length *l* is sounded together with another organ pipe of length *l* + x in their fundamental tones (x << *l*). The beat frequency heard will be (speed of sound is v):

(A) $\frac{vx}{4l^2}$ (B) $\frac{4l^2}{2x}$

(C) $\frac{vx}{2l^2}$ (D) $\frac{vx^2}{2l}$

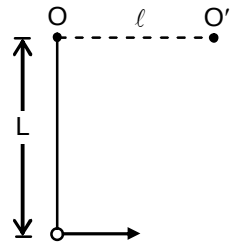
Space for Rough work

14. A rain drop of radius 1.5 mm, experiences a drag force $F = \left(2 \times 10^{-5} v\right) N$, while falling through air from a height 2km, with a velocity v. The terminal velocity of the rain drop will be nearly (use g = 10m/s^{2-})

(A) 200 m/s (B) 80 m/s

(C) 4.7 m/s (D) 3 m/s

15. A particle is hanging from a fixed pint 0 by means of a string of length L. There is a small nail O′ in the same horizontal line with O at a distance ℓ(<L) from O. The minimum velocity with which particle should be projected from its lowest position in order that it may make a complete revolution round the nail.



(A) $\sqrt{3gL}$ (B) $\sqrt{5gL}$

(C) $\sqrt{g(5L-3\ell)}$ (D) $\sqrt{g(5\ell-3L)}$

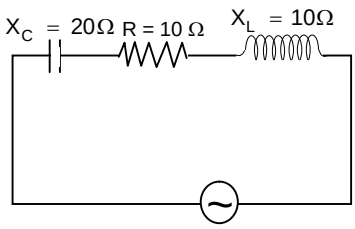
16. In LCR circuit shown in figure, then choose incorrect option

(A) Current will lead the voltage

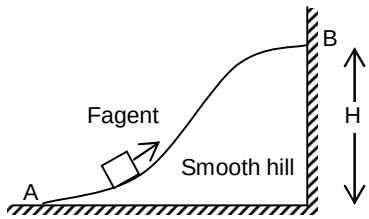
(B) rms value of current is 20 A

(C) Power factor of the circuit is $\frac{1}{\sqrt{2}}$

(D) Voltage drop across resistance = voltage drop across inductor



17. A external agent moves the block m slowly from A to B, along a smooth hill such that every time he applies the force tangentially. Find the work done by agent in this interval.

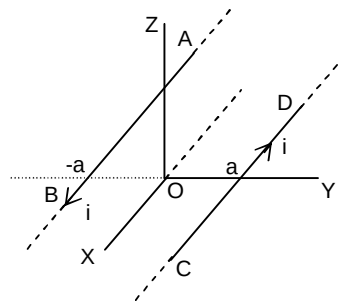


(A) $\frac{m^2g^2H^2}{L}$ (B) $\frac{mgH^2}{L}$

(C) mg(H+L) (D) mgH

Space for Rough work

18. Two long parallel wires, AB and CD, carry equal currents in opposite directions. They lie in the xy plane, parallel to the x-axis, and pass through the points (0, –1, 0) and (0, 1, 0) respectively. The resultant magnetic field is



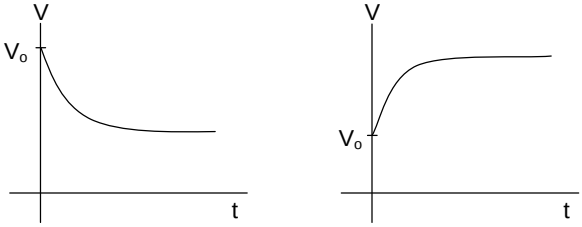
(A) zero on the x-axis
(B) maximum on the x-axis
(C) directed along the z-axis at the origin, but not at other points on the z-axis
(D) none of these

19. A tank has a hole made at its bottom. The time needed to empty the tank from level h_1 to h_2 will be proportional to

(A) $h_1 + h_2$ (B) $\sqrt{h_1} + \sqrt{h_2}$

(C) $h_1 - h_2$ (D) $\sqrt{h_1} - \sqrt{h_2}$

20. A capacitor is charged upto a potential V_o. It is then connected to a resistance R and a battery of emf E. Two possible graphs of potentials across capacitor vs time are shown.



What is the most reasonable explanation of these graphs?

(A) The first graph shows what happens when the capacitor has a lesser than E potential initially and the second shows what happens when it has a greater than E potential initially.
(B) The first graph shows what happens when the capacitor has a greater than E potential initially and the second shows what happens when it has a lesser than E potential initially.
(C) The first graph is the correct qualitative shape for any initial potential, but the second is not possible.
(D) The second graph is the correct qualitative shape for any initial potential, but the first is not possible.

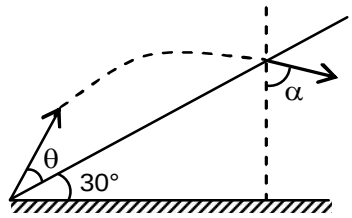
Space for Rough work

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. A particle is projected horizontally from the top of a tower with a velocity v_0. If v be its velocity at any instant, then the radius of curvature of the path of the particle at the point (where the particle is at that instant) is directly proportional to v^n, then find the value of 'n'.

22. A particle is projected at an angle $\theta = \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$ with the inclined plane of angle 30° with horizontal. If α is the angle that its velocity vector makes with vertical at the point of collision (with the inclined plane) if $\alpha = 5K°$. Find the value of 'K'.



SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. A boat which has a speed of 6 km/h in still water crosses a river of width 1 km along the shortest possible path in 20 min. The velocity of the river water in km/h is

24. A uniform solid brass sphere is rotating with angular speed ω_0 about a diameter. If its temperature is now increased by 100^0C. What will be its new angular speed (Given: $\alpha_{brass} = 2.0 \times 10^{-5} \text{per}^0\text{C}$)

25. A cube with a mass *m* = 20 *g* wettable by water floats on the surface of water. Each side of the cube has length *l* = 3 cm. The angle of contact between water and glass is zero degree and the surface tension of the water is 7.5×10^{-2} N/m. The distance between the lower face of the cube and the surface of the water is

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. Which of the following molecule experiences the lowest bond pair-bond pair repulsion?
(A) BF_3 (B) CH_4
(C) $BeCl_2$ (D) NCl_3

27. 400 mL of 0.2 M NaOH solution was mixed with 500 mL of 0.2 M HCl solution. What is the molarity of H^+ in the resulting solution?
(A) $\frac{1}{90}$ (B) $\frac{1}{45}$
(C) $\frac{2}{45}$ (D) $\frac{1}{30}$

28. If the average kinetic energy of N_2 gas is x J at a certain temperature, what will be the average kinetic energy of N gas in J at the same temperature?
(A) 2x (B) $\frac{x}{2}$
(C) 4x (D) x

29. $X(g) + 2Y(g) \longrightarrow Products$
The rate equation of above reaction is given as follows:
Rate = $k[X][Y]^{0.5}$
What is the unit of rate constant of the reaction?
(A) $mol^{1/2}L^{-1/2}s^{-1}$ (B) $mol^{-1/2}L^{1/2}s^{-1}$
(C) $mol^{3/2}L^{-3/2}s^{-1}$ (D) $mol^{-3/2}L^{3/2}s^{-1}$

Space for Rough work

30. $A(s) \rightleftharpoons 2B(g) + C(\ell)$

What will be the equilibrium partial pressure of 'B' in terms of the equilibrium constant K_P?

(A) $\sqrt{K_P}$ (B) $(K_P)^2$

(C) $K_P/2$ (D) $2K_P$

31. The strongest $p\pi - d\pi$ back bonding is observed in:

(A) CCl_4 (B) $SiCl_4$

(C) $GeCl_4$ (D) $SnCl_4$

32. Which of the following halide forms a ketone upon hydrolysis?

(A) $CH_3CH_2CH_2CHCl_2$ (B) $CH_3CH_2\underset{\displaystyle Cl}{\underset{|}{C}}HCH_3$

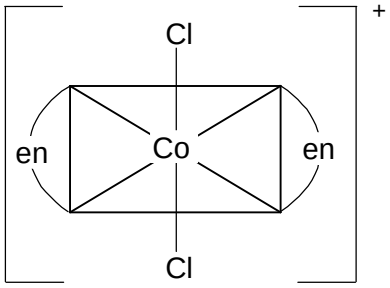
(C) $CH_3CH_2\underset{\displaystyle Cl}{\underset{|}{C}}CH_3$ (D) $CH_3CH_2\underset{\displaystyle Cl}{\underset{|}{C}}HCH_2Cl$

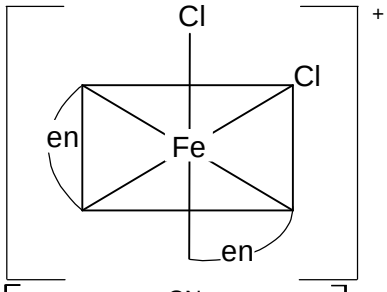
33. Which of the following aqueous solution is colourless?

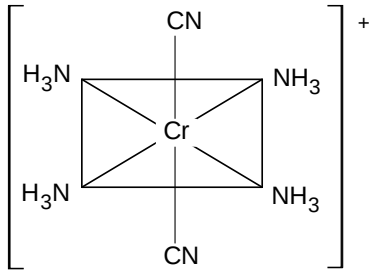
(A) $CuSO_4$ (B) $Zn(NO_3)_2$

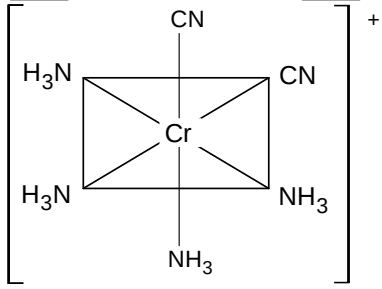
(C) $FeCl_2$ (D) $KMnO_4$

34. Which of the following complex displays optical isomerism?

(A) 

(B) 

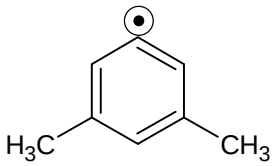
(C) 

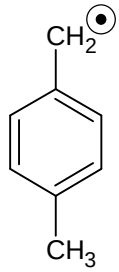
(D) 

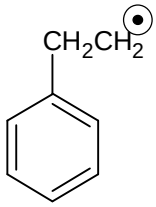
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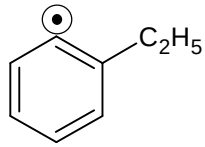
35. Which of the following interhalogen compound does NOT exist?
(A) ICl_3 (B) FCl_3
(C) IF_5 (D) IF_7

36. Which of the following free radical is most stable?

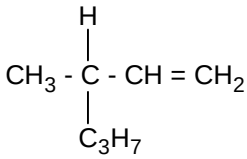
(A) 

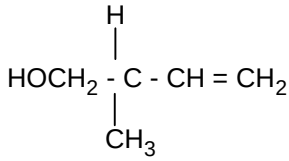
(B) 

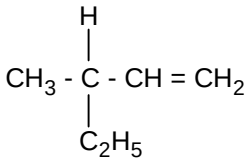
(C) 

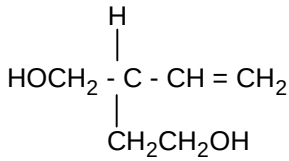
(D) 

37. Which compound loses its optical activity due to hydrogenation reaction in presence of nickel?

(A) 

(B) 

(C) 

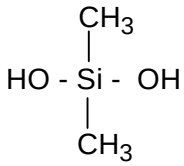
(D) 

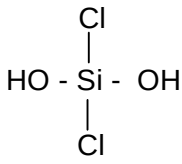
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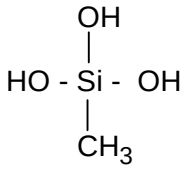
38. Which of the following substance cannot form silicate when reacts with silica(SiO_2)?
(A) NaOH (B) Na_2CO_3
(C) Na_2O (D) $NaNO_3$

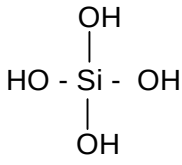
39. The compound formed by the reaction between borazine($B_3N_3H_6$) and HCl is:
(A) $B_3N_3H_7Cl$ (B) $B_3N_3H_8Cl_2$
(C) $B_3N_3H_9Cl_3$ (D) $B_3N_3H_{10}Cl_4$

40. Heating of which substance forms silica gel?

(A) 

(B) 

(C) 

(D) 

41. Which of the following is NOT a property of borax?
(A) It is soluble in water
(B) It is crystallized from aqueous solution as a decahydrated salt
(C) It's anion undergoes hydrolysis in water
(D) On heating it forms sodium boride

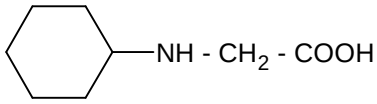
42. Which of the following group exerts electronic effects only due to sigma electrons?
(A) OH (B) CH_3
(C) NO_2 (D) CN

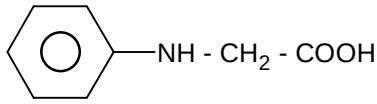
43. Which of the following compound is paramagnetic in nature?
(A) OCl_2 (B) ClO_2
(C) Cl_2O_3 (D) Cl_2O_5

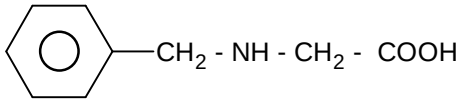
Space for Rough work

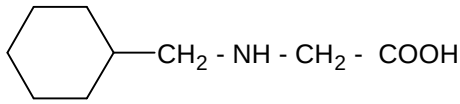
44. Which of the following is most reactive towards CH_3ONa?
(A) $CH_3COOC_2H_5$ (B) CH_3COCl
(C) CH_3CONH_2 (D) $CH_3COOCOCH_3$

45. Which of the following amino acid cannot form Zwitter ion?

(A) 

(B) 

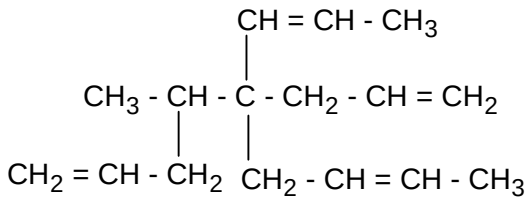
(C) 

(D) 

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. How many moles of unpaired electron(s) is/are present in a mixture containing 32 g of dioxygen (O_2) gas and 32 g of monoatomic (O) gas?

47. 

How many minimum number of moles of NBS(N-bromosuccinamide) is needed for complete reaction of one mole of the above compound in presence of sunlight?

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. The radial wave function of an atomic orbital of hydrogen atom is given below

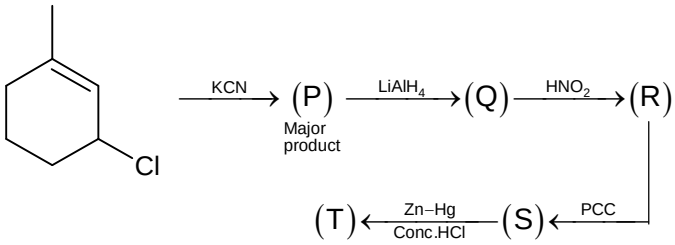
$$R_{2,1} = \frac{1}{2\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2}\left(\frac{Zr}{a_0}\right)e^{-Zr/2a_0}$$

The radial probability versus the radial distance curve of the orbital has a maxima. What is the distance of the maxima form the nucleus of the atom in $\overset{o}{A}$ unit? [$a_0 = 0.53\,\overset{o}{A}$]

49. How much heat is absorbed in joule unit by two moles of N_2 gas by heating it from 1000 K to 1315 K at constant pressure?

[R = 8.314 J K^{-1} mol^{-1}] $\left[C_P = \frac{7R}{2}\right]$

50.



What is the molar mass of (T) in g mol^{-1} unit?

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. Let $f:R \to R$, $f(x) = 8x^3 + 3x$. Then $\lim\limits_{x\to\infty} \dfrac{f^{-1}(8x) - f^{-1}(x)}{x^{1/3}}$ equals

(A) 1 (B) $1 - \dfrac{1}{3^{1/3}}$

(C) $1 - \dfrac{1}{2^{1/3}}$ (D) $\dfrac{1}{2}$

52. f is differentiable function such that $f(f(x)) = x$ where $x \in [0, 1]$. Also, $f(0) = 1$. Then the value of $\int_0^1 (x - f(x))^{2018}\, dx$ is

(A) $\dfrac{1}{2018}$ (B) $\dfrac{2}{2016}$

(C) $\dfrac{1}{2019}$ (D) $\dfrac{2}{2019}$

53. Two straight lines one being a tangent to $y^2 = 4ax$ and the other to $x^2 = 4$ by are at right angles. The locus of their point of intersection is given by

(A) $(x^2 + y^2)(ax + by) + (ay + bx)^2 = 0$ (B) $(x^2 + y^2)(ax - by) + (ay - bx)^2 = 0$

(C) $(x^2 + y^2)(ax - by) - (ay + bx)^2 = 0$ (D) $(x^2 + y^2)(ax + by) + (ay - bx)^2 = 0$

Space for Rough work

54. Let z_1, z_2, z_3, z_4 be distinct complex numbers when $|z_1| = |z_2| = |z_3| = |z_4| = 1$ Such that $0 \le \arg(z_1) < \arg(z_2) < \arg(z_3) < \arg(z_4)$ and $\arg(z_2) + \arg(z_4) - \arg(z_1) - \arg(z_3) = \pi$, then $|z_1 z_3 - (z_1 - z_3)(z_2 - z_4)|^2 - |(z_1 - z_3)(z_2 - z_4)|^2$ equals _____

(A) 1 (B) 2
(C) 3 (D) 4

55. Let M be a 3×3 matrix with all entries being 0 or 1. Then all possible values for $\det(M)$ are

(A) $\{0, \pm 1\}$ (B) $\{0, \pm 1, \pm 2\}$
(C) $\{0, \pm 1, \pm 3\}$ (D) $\{0, \pm 1, \pm 2, \pm 3\}$

56. Suppose Bunty has 4 identical green tennis balls and 5 identical red tennis balls. In how many ways can Bunty arrange these 9 balls so that no two green balls are together and no three red balls are together.

(A) 8 (B) 9
(C) 11 (D) 12

57. Let $f(x) = \frac{1}{2} x \sin x - (1 - \cos x)$. The smallest positive integer k such that $\lim_{x \to 0} \frac{f(x)}{x^k}$ exists finitely and is non zero is

(A) 3 (B) 4
(C) 5 (D) 6

Space for Rough work

58. The eccentricity of conic given by the equation $\left(x\tan10^o + y\tan20^o + \tan30^o\right)\left(x\tan120^o + y\tan220^o + \tan320^o\right) + 2019 = 0$ is

(A) $e = \sqrt{3}$ (B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2}$

59. Let $f:(0,\infty) \to R$ be defined by $f(x) = \lim_{n\to\infty} \cos^n\left(\frac{1}{n^x}\right)$. Let $x = c$ be the point where f(x) is discontinuous. Then f(c) =

(A) 0 (B) $e^{1/2}$

(C) 1 (D) $e^{-1/2}$

60. The integral $\int \frac{3x^4 + 2x^3 - 2x + 1}{x^5 + x + 1} dx$ equals

(A) $\frac{2}{3}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + \tan^{-1}\left(x^3 - x^2 + 1\right) + c$ (B) $\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + \ell n\left|x^3 - x^2 + 1\right| + c$

(C) $\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right) + \tan^{-1}\left(x^3 - x^2 + 1\right) + c$ (D) $\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right) + \ell n\left|x^3 - x^2 + 1\right| + c$

61. $\lim_{n\to\infty} \frac{1}{n} \sum_{0\le i\le j\le n} \sum \frac{1}{\sqrt{n^2 + (i+j)n + ij}}$ equal

(A) $2\left(3 + 2\sqrt{2}\right)$ (B) $3\left(2 + 3\sqrt{3}\right)$

(C) $2\left(3 - 2\sqrt{2}\right)$ (D) $3\left(2 - 3\sqrt{3}\right)$

62. If $S_n = {}^nC_0 - {}^{n-1}C_1 + {}^{n-2}C_2 - {}^{n-3}C_3 + ...$ then which of the following is always true

(A) $S_n = 0$ (B) $S_n = 1$

(C) $S_n = -1$ (D) S_n is periodic with period 6

Space for Rough work

63. Suppose 36 objects are placed along a circle at equal distances. In how many ways can 3 objects be chosen from amongst them so that no two of the three chosen objects are adjacent nor diametrically opposite.
(A) 5002 (B) 5400
(C) 5412 (D) 5500

64. Find all values of 'a' so that the equation $x^2 + (3 - 2a)x + a = 0$ has exactly one root on $(-1, 2)$.
(A) $a \in \left(-\infty, \frac{2}{3}\right) \cup \left(\frac{10}{3}, \infty\right)$ (B) $a \in \left(-\infty, \frac{2}{3}\right] \cup \left[\frac{10}{3}, \infty\right)$
(C) $a \in \left(\frac{2}{3}, \frac{10}{3}\right)$ (D) $a \in \left[\frac{2}{5}, \frac{10}{3}\right]$

65. Suppose the sum of the first m – terms of an A.P. is n and the sum of the first n – terms is m. Than the sum of first $(m + n)$ terms is
(A) 1 – mn (B) mn – 5
(C) – (m + n) (D) m + n

66. A parking lot has 16 spaces in a row. Twelve cars arrive, each of which required one parking space, and their drivers choose spaces at random from among the available spaces. Uncle Sam then arrives in his mini – truck, which requires two adjacent parking spaces. What is the probability that Uncle Sam is able to park his min – truck.
(A) $\frac{13}{28}$ (B) $\frac{17}{28}$
(C) $\frac{11}{28}$ (D) $\frac{1}{20}$

Space for Rough work

67. $f:[0,8]\to[0,8]$ be twice differentiable function with continuous and positive first derivative such that $f(0)=0; f(4)=1; f(8)=1$, then

(P) There exist some $c\in(0,8)$ where $f'(c_1)=\frac{1}{4}$

(Q) There exist some $c\in(0,8)$ where $f'(c)=\frac{1}{12}$

(R) There exist $c_1,c_2\in[0,8]$ where $8f'(c_1)f(c_2)=1$

(S) There exist some $c\in[0,8]$ such that $f(f(8))-f(f(0))=8(f'(c))^2$

Then which of the following is correct

(A) P, Q are correct
(B) P, Q, R are correct
(C) P, R are correct
(D) P, Q, R, S are correct

68. Equation of three lines $\vec{r}=\mu(\hat{i}+\hat{j}); \vec{r}=\mu(\hat{i}+\hat{j}); \vec{r}=r(\hat{i}+\hat{j}+\hat{k})$ and a plane x + y + z = 1 are given them area of triangle formed by point of intersection of line and plane is Δ, then $(6\Delta)^2$ equals

(A) 0.75 (B) 0.25
(C) 0.50 (D) 0.40

69. An ellipse touches X – axis at O(0, 0) and has its centre at (3, 4). It is known that slope of major axis is 1. The eccentricity of the ellipse is

(A) $\sqrt{\frac{3}{4}}$ (B) $\sqrt{\frac{4}{5}}$
(C) $\sqrt{\frac{5}{6}}$ (D) $\sqrt{\frac{6}{7}}$

70. The negation of $(\sim p\wedge q)\vee(p\wedge\sim q)$ is

(A) $(p\vee\sim q)\vee(\sim p\vee q)$ (B) $(p\vee\sim q)\wedge(\sim p\vee q)$
(C) $(p\wedge\sim q)\wedge(\sim p\vee q)$ (D) $(p\wedge\sim q)\wedge(p\vee\sim q)$

Space for Rough work

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. The value of $a \in R$, such that $f:R \to R$ defined by $f(x) = x^3 + x^2 + a\cos x$ is injective

72. $S = \frac{1}{2} + \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{5}{32} + \frac{8}{64} + \frac{13}{128} + ...\infty =$

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. Point 'O' is the centre of the ellipse with major axis AB and minor axis CD. Point F is one focus of the ellipse. If OF = 6 and the diameter of the inscribed circle of ΔOCF is 2, then the product (AB) (CD) is $\sqrt{(AB)(CD)-1}$ equals

74. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors satisfying $|\vec{a}-\vec{b}|^2 + |\vec{b}-\vec{c}|^2 + |\vec{c}-\vec{a}|^2 = 9$, then value of $|2\vec{a} + 5\vec{b} + 5\vec{c}|$ is _____

75. If $\sum_{i=1}^{9}(x_i - 5) = 9$ and $\sum_{i=1}^{9}(x_i - 5)^2 = 45$, then the standard deviation of the 9 items $x_1, x_2, ...x_9$ is

Space for Rough work